

Mastering KNIME Analytics Platform

A Comprehensive Guide to Visual Programming

(Focus on Version 5.x Modern UI)

Abdul Rehman

1. Introduction

Understanding the Platform &
Ecosystem

What is KNIME?

Visual Data Science

KNIME (Konstanz Information Miner) is a free and open-source data analytics, reporting, and integration platform.

- **No-Code/Low-Code:** Build workflows by dragging and dropping nodes.
- **End-to-End:** Covers everything from data blending to deployment.

Key Philosophy

KNIME is built on the concept of **Visual Programming**.

Instead of writing lines of code, you connect steps (nodes) into a pipeline (workflow).

This makes the data process transparent, documented, and reproducible.

Why Choose KNIME?



Visual

Intuitive graphical interface makes logic easy to follow and debug compared to script files.



Open Source

Free to use with no limitations on data size or node usage. Extensive community support.



Integrative

Acts as a glue, connecting different tools (Python, R, SQL, Tableau, PowerBI) in one workflow.

Installation & Workspaces

Getting Started

- Download the installer from knime.com/downloads.
- Available for Windows, Mac (Intel/Apple Silicon), and Linux.
- No license key required for the Analytics Platform.

The Workspace

When you launch KNIME, you select a **Workspace**. This is a folder on your computer where all your workflows, data, and settings will be stored.

Tip: Keep your workspace organized with logical subfolders (e.g., Data, Projects, Archive).

The Modern UI

Clean & Intuitive

KNIME 5.x introduces a sleek, modern user interface designed for ease of use.

It reduces clutter and focuses on the workflow canvas. The search capabilities are significantly improved, and common actions are accessible via clear buttons.

KNIME Analytics Platform

Use Case ×

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```
graph LR; ExcelReader[Excel Reader  
Reading CeltsData.xls] --> Metanode[Metanode]; CSVReader[CSV Reader  
Reading ContractData.csv] --> Metanode; Metanode --> KMeans[k-Means  
10 clusters on all numerical inputs]; KMeans --> Denormalizer1[Denormalizer (PMML)  
Back to original data range]; KMeans --> Denormalizer2[Denormalizer (PMML)  
Back to original data range];
```

► 1: File Table ⚙ Flow Variables

Rows: 100 of 3333 Columns: 16

ID	VMail Message Number (Integer)	Day Mins Number (double)	Eve Mins Number (double)	Night Mins Number (double)	Intl Mins Number (double)	CustServ Calls Number (Integer)	Day Calls Number (Integer)	Day Charge Number (double)	Eve Calls Number (Integer)	Eve Charge Number (double)	Night Calls Number (Integer)	Night Charge Number (double)
Row0	25	265.1	197.4	244.7	10.0	1	110	45.07	99	16.78	91	11.0
Row1	26	161.6	195.5	254.4	13.7	1	123	27.47	103	16.62	103	11.4
Row2	0	243.4	121.2	162.6	12.2	0	114	41.38	110	10.3	104	7.32
Row3	0	299.4	61.9	196.9	6.6	2	71	50.9	88	5.26	89	8.86
Row4	0	166.7	148.3	186.9	10.1	3	113	28.34	122	12.61	121	8.41
Row5	0	223.4	220.6	203.9	6.3	0	98	37.98	101	18.75	118	9.18
Row6	24	218.2	348.5	212.6	7.5	3	88	37.09	108	29.62	118	9.57

Classic vs. Modern UI

Classic UI (v4.x style)

- Eclipse-based interface with many panels visible at once.
- Preferred by long-time power users for deep customization.
- "KNIME Explorer" view for file navigation.

Modern UI (v5.x style)

- Browser-based look and feel.
- "Space Explorer" replaces KNIME Explorer.
- Simplified node repository with tags.
- You can toggle between them using the "Open in Classic UI" button top right.

Key UI Elements



Canvas

The large central area where you drag, drop, and connect nodes to build your workflow.



Node Repository

The library of all available nodes. In Modern UI, it's a searchable panel on the left.



Description

A dynamic panel that shows help and configuration details for the currently selected node.

2. Core Concepts

Nodes, Workflows, and Data

Flow

The Node Concept

What is a Node?

A node is the basic processing unit in KNIME. It performs a single, specific task.

- > **Input:** Receives data or a model.
- > **Configuration:** User settings (dialog).
- > **Execution:** Performs the task.
- > **Output:** Passes result to the next node.

Visual Representation

Nodes are represented as colored boxes with icons.

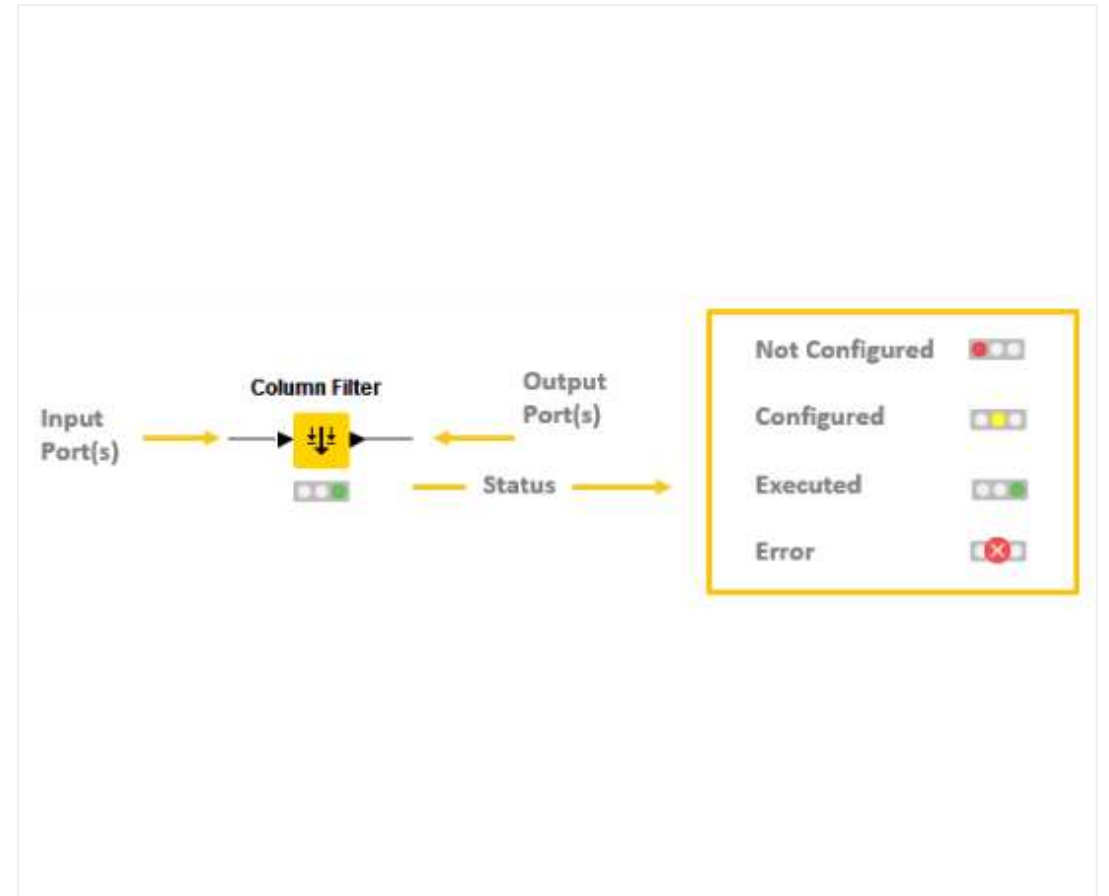
- > **Orange:** Read/Write data.
- > **Yellow:** Transformation/Manipulation.
- > **Blue:** Visualization.
- > **Green:** Machine Learning.

Traffic Light System

Status Indicators

Every node has a status light under it indicating its state:

- > **Red:** Not configured or has an error.
- > **Yellow:** Configured and ready to execute.
- > **Green:** Executed successfully. Data is available at output.



Understanding Ports

Data Ports (Triangles)

- **Black Triangles:** Standard data tables.
- Input ports are on the left, output ports on the right.
- You can inspect the data by hovering over the port (Modern UI) or right-clicking (Classic).

Special Ports (Squares)

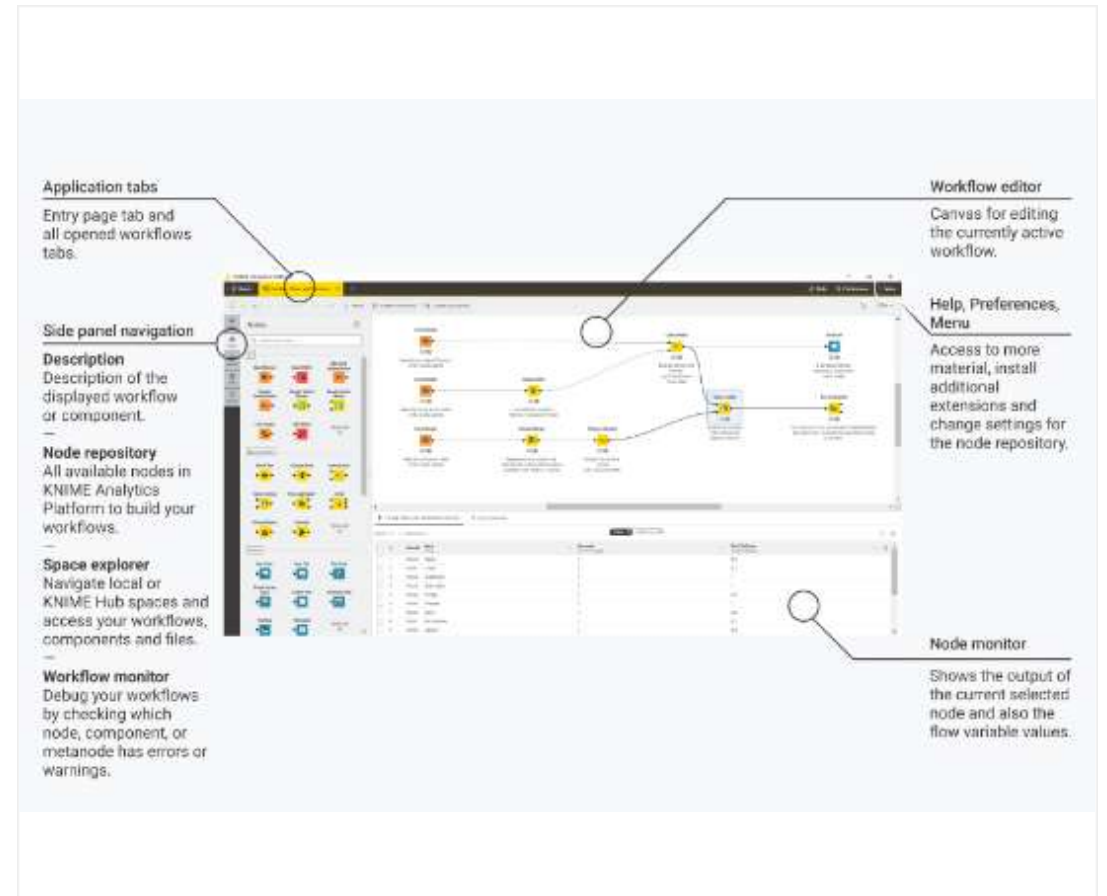
- **Blue Squares:** PMML Models (Machine Learning models).
- **Red Circles (Hidden):** Flow Variables (for controlling workflow logic, not data).
- **Dark Blue Squares:** Database Connections.

Node Repository

Finding Tools

The repository contains thousands of nodes categorized by function (IO, Manipulation, Analytics, etc.).

- > **Search:** Use the search bar at the top to find nodes by name (e.g., "Excel Reader").
- > **Drag & Drop:** Simply drag a node from the repository onto the canvas to use it.
- > **Fuzzy Search:** KNIME search is smart enough to find "Group" if you type "Agg".



Workflow Editor

Connecting Nodes

To create a flow, click the output port of one node and drag the connection line to the input port of another node.

If the connection is valid (compatible types), the line will snap into place.

Organization

- > **Annotations:** You can draw colored boxes around groups of nodes to document your work.
- > **Alignment:** Use the toolbar buttons to align nodes horizontally or vertically for a clean layout.

3. Data Access

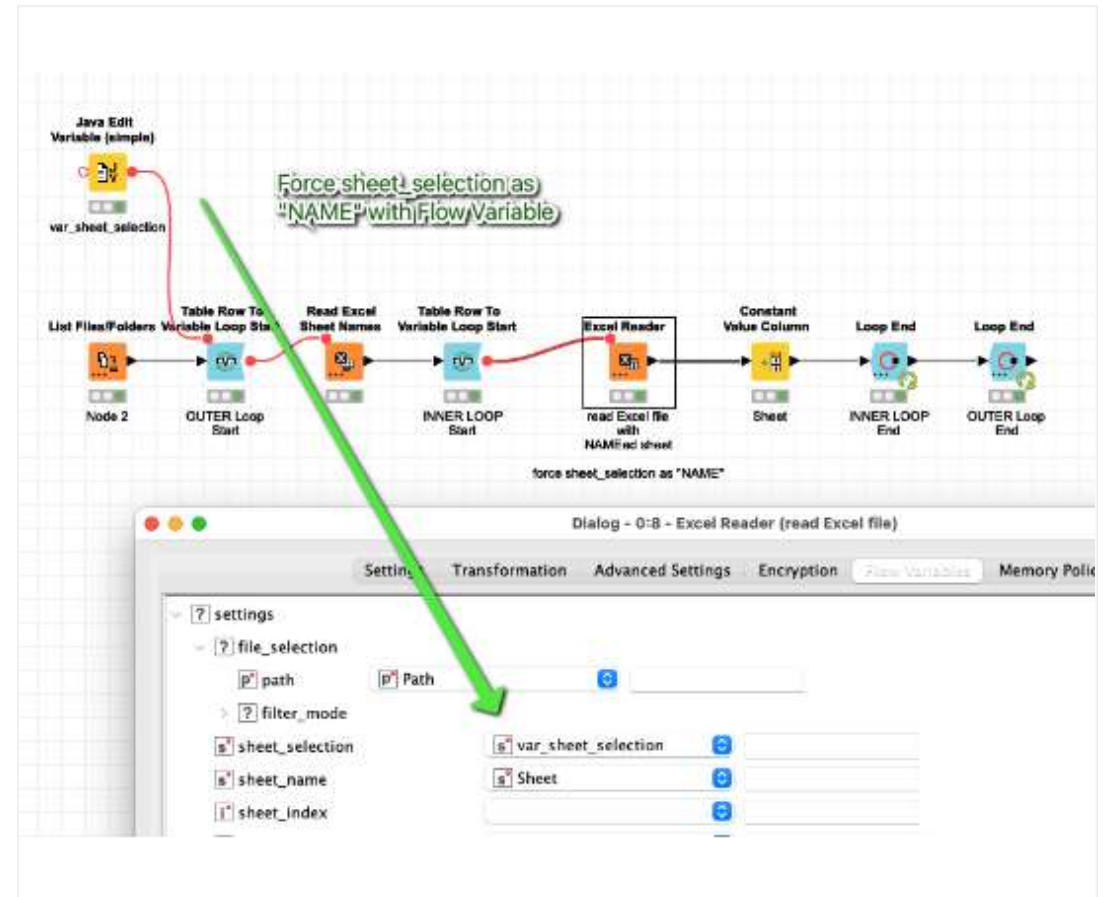
Reading Files, Databases, and APIs

Reading Excel Files

Excel Reader Node

Reads .xls and .xlsx files.

- > **Configuration:** Select the file path.
- > **Sheet Selection:** Choose specific sheets to read.
- > **Preview:** The bottom of the dialog shows a preview of the data structure.
- > Supports reading from local disk or KNIME Hub/Server.



Reading Text/CSV

CSV Reader / File Reader

Used for importing flat files (Comma Separated Values, Tab Delimited, etc.).

- **Autodetection:** KNIME attempts to guess the delimiter and column types.
- **Fine Tuning:** You can manually set the delimiter, quote character, and comment character.

Table Reader

Reads the internal .table format.

This is the fastest way to read data in KNIME because the file is already optimized and indexed for the platform.

Database Connectivity



DB Connector

Establishes the connection using JDBC drivers (MySQL, PostgreSQL, Oracle, SQLite, Snowflake, etc.).



DB Table Selector

Selects a table or writes a SQL query to fetch data without downloading it yet.



DB Reader

Executes the query and downloads the result into KNIME as a local data table.

The KNIME Protocol

Relative Paths

Avoid using hardcoded paths like
C:\Users\John\Desktop.... This breaks the workflow
when shared.
Use the **knime://** protocol instead.

Examples

- `knime://knime.workflow/data/file.csv`
(Looks in a folder named 'data' inside the workflow folder)
- `knime://knime.mountpoint/shared/data.xlsx`
(Looks in the workspace root)

4. Data Manipulation

ETL: Filtering, Transforming, and Aggregating

Data Inspection

The Table View

At any point in the workflow, you can inspect the data.

- **Modern UI:** A table preview appears at the bottom "Node Monitor" panel when a node is selected.
- **Classic UI:** Right-click a node and select the last option (e.g., "Output Table").

Statistics

The "Data Explorer" or "Statistics" tab in views provides distributions, min/max values, and missing value counts for every column instantly.

Row Filtering

Row Filter Node

The workhorse for subsetting data.

- > **Attribute Filtering:** Keep rows where "Region" is "West".
- > **Range Filtering:** Keep rows where "Sales" > 500.
- > **Pattern Matching:** Use wildcards (*, ?) or RegEx.

Modern Dialog

The new v5.x dialog allows you to add multiple conditions in a single node easily, mimicking a SQL WHERE clause visually.

Column Filtering

Column Filter Node

Removes unnecessary columns to reduce memory usage and clutter.

- > **Manual Selection:** Drag columns from "Include" to "Exclude".
- > **Type Selection:** e.g., "Exclude all String columns".
- > **Name Selection:** Keep columns that start with "2023_".

Best Practice

Filter columns as early as possible in your workflow. Processing 10 columns is much faster than processing 100.

Handling Missing Values

The Issue

Missing data (represented as ? in red) can cause errors in calculations or ML models.

Missing Value Node

Provides strategies to handle gaps:

- **Remove Row:** Delete the entire record.
- **Impute (Number):** Replace with Mean, Median, or Fixed Value (e.g., 0).
- **Impute (String):** Replace with "Unknown" or Mode (Most Frequent).

Sorting Data

Sorter Node

Reorders the rows based on one or more columns.

- Ascending (A-Z, 0-9)
- Descending (Z-A, 9-0)

Multi-Level Sort

You can add multiple sort keys. For example, first sort by "Country" (Asc), and then within each country, sort by "Sales" (Desc).

String Manipulation

String Manipulation Node

The classic powerhouse for text cleaning.

- `toUpperCase() / lowercase()`
- `join(str1, str2)`
- `replace(str, search, replace)`
- `regexReplace()`

Usage

It works on a single column at a time (mostly) and creates a new column or replaces the existing one.

The Expression Node



New in v5.3

Combines String Manipulation, Math Formula, and Column Expressions into one modern interface.



AI Assistant

Includes an embedded AI chat. You can type "Extract the email domain" and it will write the RegEx code for you.



Multi-Column

Can handle multiple calculations in a single node, reducing workflow clutter.

Rule Engine

Conditional Logic

Think of this as a supercharged "If-Then-Else" tool.

It assigns a value to a new column based on a set of rules defined in a table.

Syntax Example

```
$Sales$ > 1000 => "High Priority"  
$Region$ LIKE "West" => "Western Team"  
TRUE => "Standard"
```

(The first rule that matches wins)

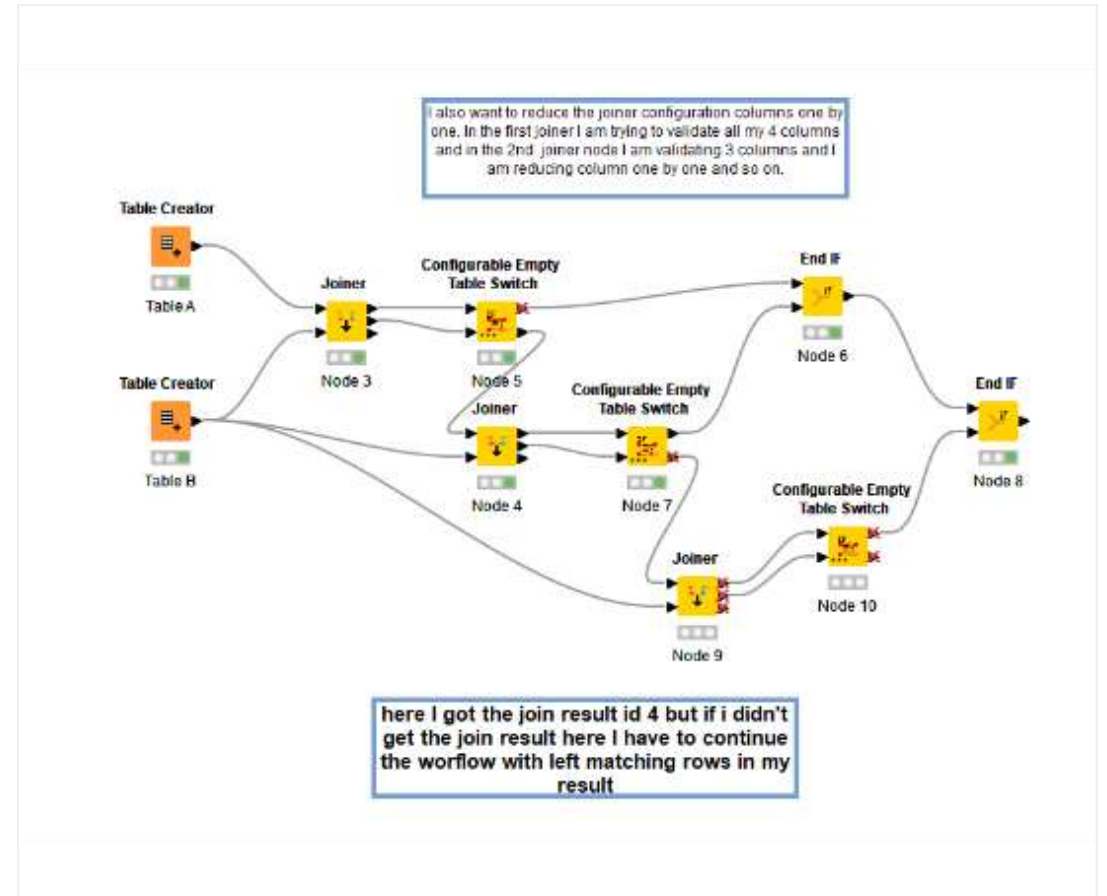
Joining Data

Joiner Node

Combines columns from two tables based on a common key (ID).

- Inner Join: Only matching rows.
- Left/Right Outer: Keep unmatched rows from one side.
- Full Outer: Keep everything.

The modern dialog makes column selection much easier.



Concatenation

Concatenate Node

Stacks tables on top of each other (Union).

Useful when you have data split by month (Jan.csv, Feb.csv) and want one master table.

Intersection vs Union

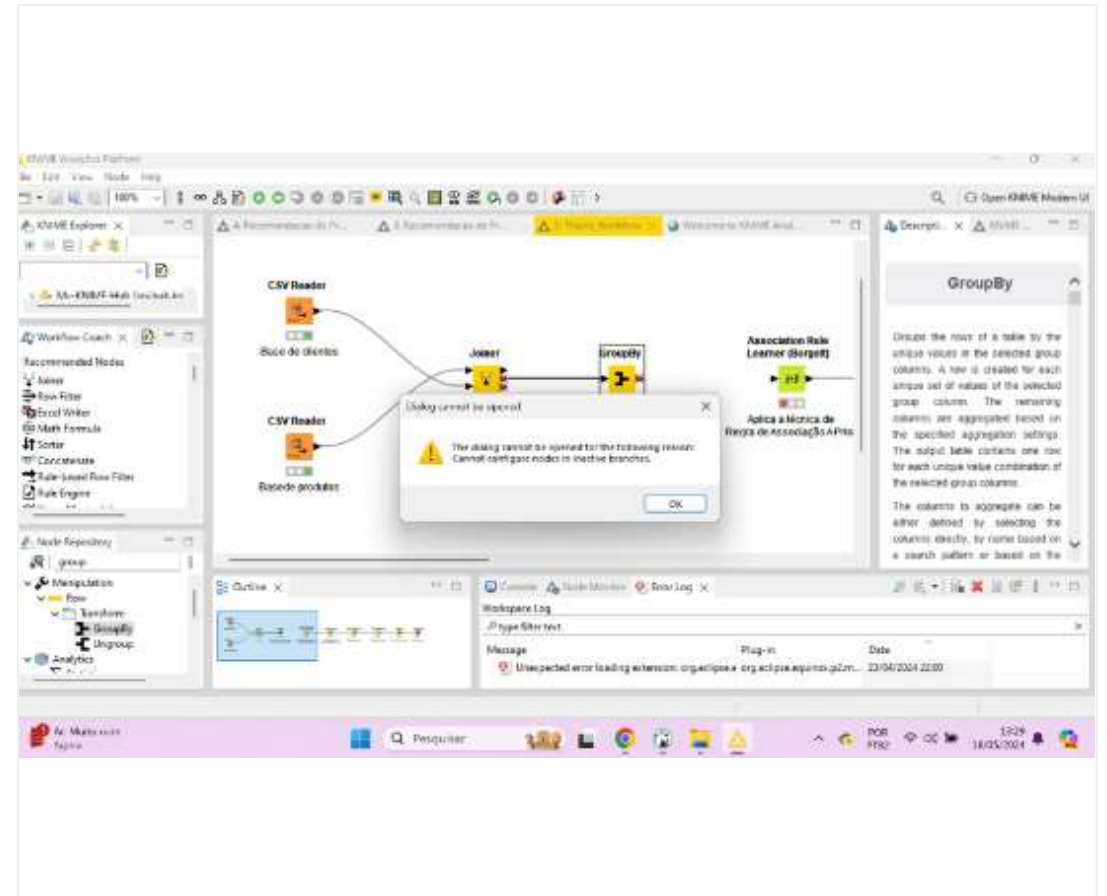
- > **Union of Columns:** Output contains all columns from inputs. Missing columns in one input are filled with ?.
- > **Intersection of Columns:** Output only contains columns that exist in BOTH inputs.

Aggregation (GroupBy)

GroupBy Node

Similar to SQL GROUP BY or Excel Pivot Tables.

- **Group Columns:** The categorical dimensions (e.g., Region, Product).
- **Aggregation Columns:** The numbers to summarize (e.g., Sum(Sales), Mean(Profit)).



Pivoting & Unpivoting

Pivoting (Long to Wide)

Turns unique values in a column into new column headers.

Example: Turning a "Month" column into 12 columns (Jan, Feb...).

Unpivoting (Wide to Long)

The reverse process. It collapses multiple columns into Key-Value pairs.

Essential for preparing data for visualization tools like Tableau or PowerBI.

5. Visualization

Charts, Views, and Dashboards

Bar Charts

Bar Chart Node

KNIME uses a JavaScript-based visualization library (ECharts in modern UI).

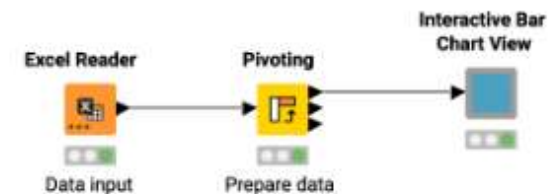
- > Interactive: Zoom, pan, and select.
- > Configurable titles, axes labels, and tooltips.
- > Supports stacked and grouped bars.

Interactive Bar Chart Example

This workflow demonstrates how to create interactive bar chart using Refresh Widget node.

Forum link:

<https://forum.knime.com/t/grouped-bar-chart/33602>



Scatter Plots

Visualizing Relationships

The Scatter Plot node plots two numerical variables against each other.

It is crucial for identifying correlations (positive/negative), clusters, or outliers in your dataset.

Interactivity

In the interactive view, selecting points in the scatter plot can highlight the corresponding rows in a Table View if they are part of the same component.

Color Manager

Assigning Colors

The Color Manager node assigns colors to rows based on a column's value.

- **Nominal (Categorical):** Assign distinct colors (e.g., Red for "Fail", Green for "Pass").
- **Numeric (Continuous):** Assign a gradient (e.g., Blue to Red for Temperature).

Persistence

Once assigned, the color persists through the workflow. Subsequent plots (Scatter, Bar) will inherit these colors automatically.

Composite Views (Dashboards)

Components

If you wrap multiple visualization nodes into a **Component**, KNIME combines their individual views into a single dashboard layout.

Layout Editor

Inside the component, you can use the "Visual Layout" editor to arrange charts, add text, and create a fully functional web-application style dashboard.

6. Machine Learning

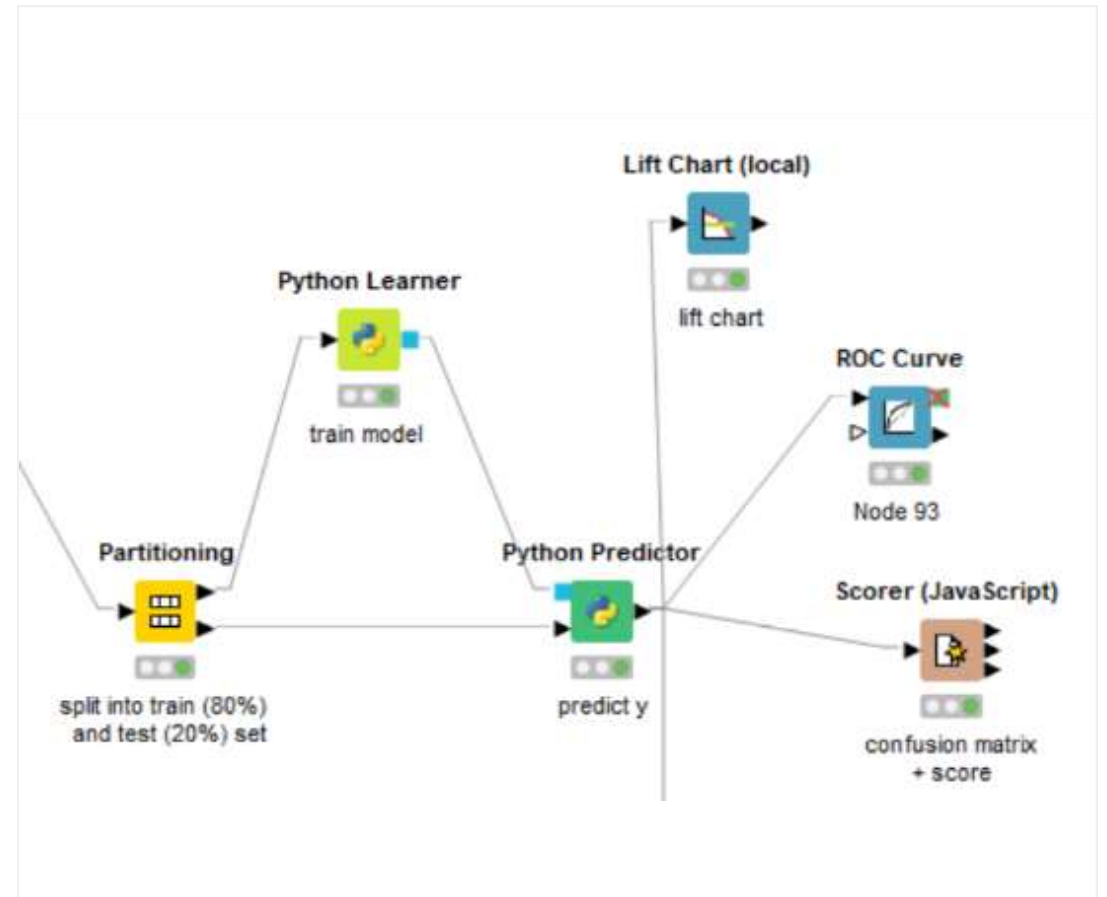
Training, Testing, and Scoring Models

The Learner/Predictor Paradigm

Two-Step Process

KNIME splits ML into two distinct nodes:

1. **Learner Node:** Takes training data and outputs a model (Blue Square port).
2. **Predictor Node:** Takes the model and new data to generate predictions.



Partitioning Data

Partitioning Node

Essential for validating models. It splits the dataset into two subsets.

- **Training Set (e.g., 70%):** Used to build the model.
- **Test Set (e.g., 30%):** Used to evaluate how well the model performs on unseen data.

Stratified Sampling

Ensures that the class distribution (e.g., Yes/No churn) remains consistent in both splits, preventing bias.

Scoring & Evaluation

Scorer Node

Compares the "Actual Class" vs. "Predicted Class" to evaluate performance.

- > **Confusion Matrix:** Shows True Positives, False Positives, etc.
- > **Accuracy Statistics:** Precision, Recall, F1-Score, Cohen's Kappa.

For numeric models (Regression), use the **Numeric Scorer** (R^2 , RMSE).

	Predicted class POSITIVE (spam 📧)	Predicted class NEGATIVE (ad 👤)	Predicted class NEGATIVE (normal 👤)
Actual class POSITIVE (spam 📧)	TRUE POSITIVES 📧 📧 27	FALSE NEGATIVES 📧 👤 286 📧 👤 40	
Actual class NEGATIVE (ad 📧)	FALSE POSITIVES 📧 📧 1	TRUE NEGATIVES 📧 👤 37 📧 👤 9	
Actual class NEGATIVE (normal 📧)	FALSE POSITIVES 📧 📧 5	TRUE NEGATIVES 📧 👤 16	TRUE NEGATIVES 📧 👤 500

7. Advanced & Control

Loops, Automation, and Extensions

Workflow Control

Loops

Iterate over a set of files, columns, or rows.

- **Loop Start:** Defines the start.
- **Loop End:** Collects the results from each iteration.

Switches

Case Switch / If-Switch: Directs the workflow path based on a condition (e.g., IF data is empty, skip processing).

Components vs. Metanodes

Metanodes (Grey)

Purely for visual organization. They collapse a chunk of nodes into a single folder-like node to clean up the canvas. They don't affect variable scope.

Components (Blue)

Like "Sub-Workflows". They encapsulate logic and isolate flow variables. They can also have their own configuration dialogs and interactive views (Dashboards).